GATE-AII BRANCHES
Engineering Mathematics



Probability and Statistics

Lecture No.- 13



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Recap of Previous Lecture



Topic Concept of probability distribution

bernoulli Distribution

Topic Concept of Binomial distribution

Topic Problems based on Binomial distribution

Topics to be covered



Topic Problems based on Binomial distribution

Topic Problems based on Gaussian distribution

Bernoulli Distribution

Single Trial < S, F

$P[X=x] = p^x (1-p)^{1-x}$

$E[X] = p$

$V(X) = pq$

$S.D = \sqrt{pq}$

binomial Distribution

$P[X=r] = {}^nC_r\, p^r q^{n-r}$

✓ Trials fixed
✓ P – given
✓ r = No. of success

$E[X] = np$

$V(X) = npq$

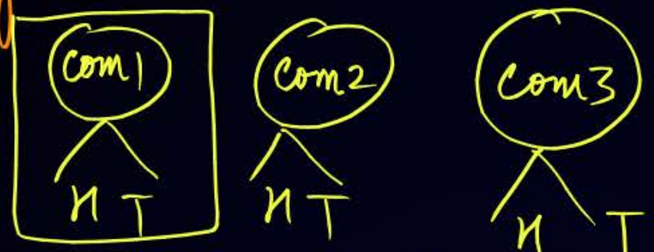
$S.D = \sqrt{npq}$

$B(p) \xrightarrow{\text{large NO}} B(n, p)$

Bernoulli — Binomial

fundamentals

Tossing A THREE coins



- ✓ P(atleast one HEAD)
- ✓ P(exactly one HEAD)
- ✓ P(Almost Two HEAD)

Arrival

Distribution Function (Discrete)

Coin $\leftarrow$ H, T

Bernoulli – Trials $\leftarrow$ success, Failure

$P(H) = \frac{1}{2}$ $P(T) = \frac{1}{2}$

$\checkmark p = \frac{1}{2}$ $q = \frac{1}{2}$ $n = 3$

Binomial Distribution $B(n, p)$



$P(\text{atleast one HEAD}) \Rightarrow$

$= P(X=1) + P(X=2) + P(X=3)$



$P(X \geq 1) = P(\text{at least one HEAD}) = P(X=1) + P(X=2) + P(X=3)$

1H, 2H, 3H — Add

$P(X=r) = {}^{n}C_r\, p^r q^{n-r}$

$p = q = \frac{1}{2}, \quad n = 3$

$P(X=1) = {}^{3}C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^{3-1}$ +

$P(X=2) = {}^{3}C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^{3-2}$ +

$P(X=3) = {}^{3}C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^{3-3}$

$P(X \geq 1) = {}^{3}C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^2 + {}^{3}C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^1 + {}^{3}C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^0$

$P(X \geq 1) = \frac{7}{8}$

0	1	2	3	4

$P(X=0) + P(X=1) + P(X=2) + \cdots + P(X=r) = 1$

$$P(X \geq 1) = 1 - P(X=0)$$

$$= 1 - {}^3C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{3-0}$$

$$= 1 - \frac{1}{8} = \frac{7}{8}$$ ✓

$${}^nC_r = \frac{n!}{(n-r)!\,r!}$$

3) $P(\text{at most Two HEAD}) = P(X=0) + P(X=1) + P(X=2)$

max

$$= {}^3C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{3-0} + {}^3C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^{3-1} + {}^3C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^{3-2}$$

max (2) HEAD

$\Rightarrow$ 

0 HEAD
1 HEAD +
2 HEAD +

(2) $P(\text{exactly one HEAD}) = P(X=1)$

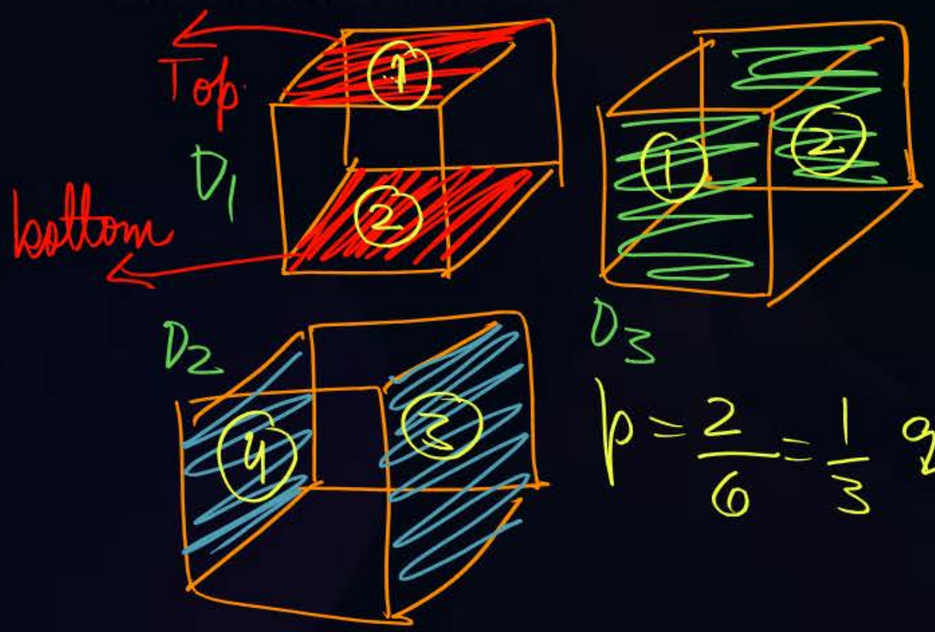
$$\Rightarrow {}^3C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^{3-1} =$$



Topic :Probability and statistics

NCERT Vol-2

#Q. Consider and unbiased cubic die with opposite faces coloured identically and each face colored red, blue or green such that each colour appears only two times on the die. If the die is thrown thrice, the probability of obtaining red colour on top face of the die at least twice is-



success

$n = 3 \quad P(X \geq 2)$

p = success = red color

q = failure = Not red color

$p = \frac{1}{3}$

$q = \frac{2}{3}$

$p = \frac{2}{6} = \frac{1}{3} \quad q = \frac{4}{6} = \frac{2}{3}$

Binomial Distribution

$p = \frac{1}{3}$ $q = \frac{2}{3}$ $x \geq 2$ $n = 3$

Using Binomial Distribution

$P(X = x) = {}^{n}C_{x}\, p^{x} q^{n-x}$

$P(X \geq 2) = P(X = 2) + P(X = 3)$

$= {}^{3}C_{2} \left(\frac{1}{3}\right)^{2} \left(\frac{2}{3}\right)^{3-2} + {}^{3}C_{3} \left(\frac{1}{3}\right)^{3} \left(\frac{2}{3}\right)^{3-3}$

$$\boxed{P(X \geq 2) = \frac{7}{27}}$$



Topic : Probability and statistics

$P(x \geq 2)$

#Q. The probability of obtaining at least two six in throwing a fair dice 4 times is-

$P(\text{success}) = \frac{1}{6}$ $n = 4$

$P(\text{failure}) = \frac{5}{6}$

success = 6

Failure = 6

$n = 4$

A 425/432

B 19/144

C 13/144

D 125/432

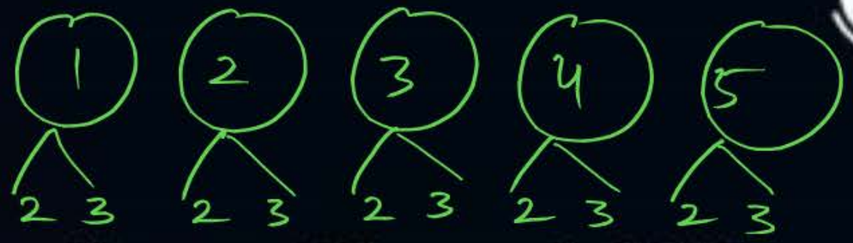
$P(x \geq 2) = P(x=2) + P(x=3) + P(x=4)$

$= 1 - P(x=0) - P(x=1)$

$= 1 - {}^4C_0 \left(\frac{1}{6}\right)^0 \left(\frac{5}{6}\right)^{4-0} - {}^4C_1 \left(\frac{1}{6}\right)^1 \left(\frac{5}{6}\right)^{4-1} = 0.131$

Topic :Probability and statistics





#Q. Five coins whose faces are marked 2,3 are thrown. The probability of obtaining a total of 12 is:

(2), (3) (bernoulli Trials)

Total = 12 SUCCESS.

com1 com2 com3 com4 com5

2+2+2+3+3 = 12

2 — (3 's)

3 — (2 's)

2+2+2+3+3 = 12

2 SUCCESS, 3 failure.

or

3+3+2+2+2

S F

A 5/32

B 11/16

C 5/16

D 10/16



$P(2) = \frac{1}{2} = P(S) = p$ $\quad n = 5$

$P(3) = \frac{1}{2} = P(F) = q$

$P(\text{SUM} = 12) = {}^{n}C_r\, p^r q^{n-r}$

$= {}^{5}C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^{5-3}$

✓ back 2

$= \frac{5 \times 4 \times 3}{3 \times 2 \times 1} \frac{1}{8} \times \frac{1}{2 \times 2}$

$= \frac{10}{32} = \frac{5}{16}$

${}^{5}C_3$

$= \frac{5!}{3! \times 2!}$

$= \frac{5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 1 \times 2 \times 1}$

$= 10$

$2+2+2+3+3$

SUCCESS

$r = 3$

II nd SECOND

$3+3+2+2+2$

$r = 2$

$n = 5$

$p = \frac{1}{2}$

$q = \frac{1}{2}$

$= {}^{5}C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2 = \frac{5 \times 4}{2 \times 1} \times \frac{1}{16}$

$= \frac{10}{32}$ ✓



Topic :Probability and statistics

✓ read the Problem ✗

$n = 10$ questions $p = ✓$

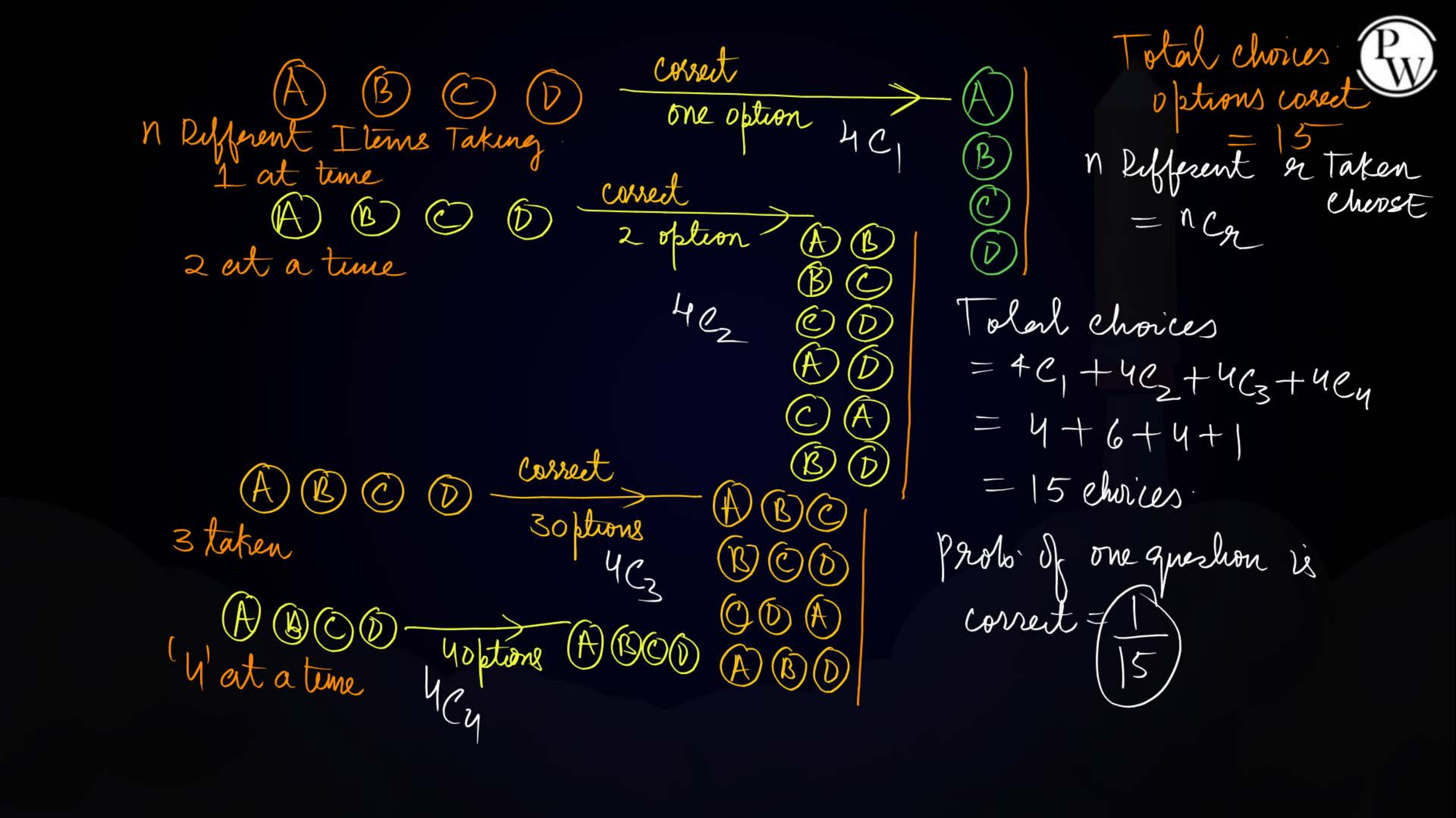
#Q. In an Examination of 10 multiple choice questions (one or more can be correct out of 4 options. A student decided to mark the answer at random.) Find the probability he gets exact two questions correct.

$r = 2$ questions

one or more can be correct

$p = ?$

I ✓
II ✓
III ✓
IIII ✓


A B C D
n Different Items Taking
1 at time
correct
one option
$^4C_1$
A
B
C
D
A B C D
2 at a time
correct
2 option
$^4C_2$
A B
B C
C D
A D
C A
B D
A B C D
3 taken
correct
3 options
$^4C_3$
A B C
B C D
C D A
A B D
A B C D
'4' at a time
4 options
$^4C_4$
A B C D
Total choices
options correct
= 15
n Different or Taken
choose
$= {}^nC_r$
Total choices
$= {}^4C_1 + {}^4C_2 + {}^4C_3 + {}^4C_4$
$= 4 + 6 + 4 + 1$
$= 15$ choices.
Prob. of one question is
correct $= \frac{1}{15}$

$$P(\text{correct}) = \frac{1}{15} \quad P(\text{Not correct}) = \frac{14}{15} = 1 - \frac{1}{15}$$

$$r = 2 \quad n = 10$$

Using binomial dist.

$$P(X = r) = {}^{n}C_{r}\, p^{r} q^{n-r}$$

$$= {}^{10}C_{2} \left(\frac{1}{15}\right)^{2} \left(\frac{14}{15}\right)^{10-2}$$

$$= {}^{10}C_{2} \times \frac{14^{8}}{15^{10}}$$



Topic :Probability and statistics

#Q. Difference between mean and variance of a binomial random variable is 1 and difference between their squares is 11. Find the probability of getting exactly three success.

$MEAN = np$ $\quad variance = npq$

$B(n, p)$

$$\begin{cases} np - npq = 1 & -(1) \\ n^2p^2 - n^2p^2q^2 = 11 & -(2) \end{cases}$$

$r = 3 \quad n = ? \quad p = ?$

$P(X = 3) =$

$np - npq = 1$

$np(1-q) = 1 \quad -①$

$n^2p^2 - n^2p^2q^2 = 11$

$n^2p^2(1-q^2) = 11 \quad -②$

$\frac{n^2p^2(1-q^2)}{np(1-q)} = \frac{11}{1}$

$\frac{n^2p^2(1-q)(1+q)}{np(1-q)} = \frac{11}{1}$

$np(1+q) = 11$

Using equⁿ ① $np(1-q) = 1$

$np = \frac{1}{1-q}$

$\frac{1}{(1-q)}(1+q) = \frac{11}{1}$

$\frac{1+q}{1-q} = \frac{11}{1}$

$1+q = 11-11q$

$1-11 = -11q-q$

$-12q = -10$

$q = \frac{10}{12} = \frac{5}{6}$

$q = \frac{5}{6}$ $\quad p = 1-q = 1-\frac{5}{6} = \frac{1}{6}$

Using equⁿ ①

$np(1-q) = 1$

$n \times \frac{1}{6}\left(1-\frac{5}{6}\right) = 1$

$n = 36$

Using $B(n, p)$

$$P(X=r) = {}^{n}C_{r}\, p^{r} q^{n-r}$$

$$= {}^{36}C_{3}\left(\frac{1}{6}\right)^{3}\left(\frac{5}{6}\right)^{36-3}$$

$$= {}^{36}C_{3}\left(\frac{1}{6}\right)^{3}\left(\frac{5}{6}\right)^{33}$$

$$= 0.81$$



Topic :Probability and statistics

fav question

#Q. An unbiased coin is tossed an infinite number of times. The probability that the fourth head appears at the tenth toss is

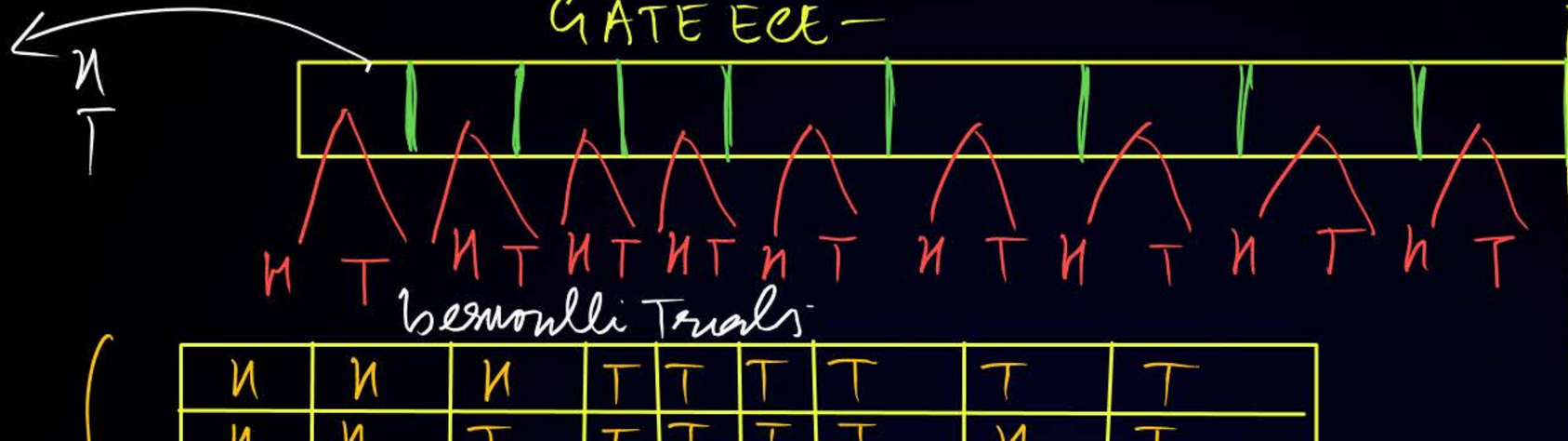
A 0.067

B 0.073

C 0.083

D 0.091

GATE ECE –



Bernoulli Trials

H	H	H	T	T	T	T	T	T
H	H	T	T	T	T	T	H	T
T	T	T	T	T	T	H	H	H
H	T	H	T	H	T	T	T	T

H H T H T H T T T

$n = 9 \quad r = 3 \quad p = \frac{1}{2} \quad q = \frac{1}{2}$

$P(X=3) = {}^9C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^{9-3}$

$P(X=3) = {}^9C_3 \frac{1}{2^9}$ ✓ Work (1)

Work 2

$P(4^{th} \text{ HEAD } 10^{th} \text{ Toss})$

$= {}^9C_3 \left(\frac{1}{2^9}\right) \times \frac{1}{2}$

$= \frac{{}^9C_3}{2^{10}}$

$= B(n,p) \times (b)$

$= 0.082$



Topic :Probability and statistics

#Q. In a manufacturing plant, the probability of making a defective bolt is 0.1. The mean and standard deviation of defective bolts in a total of 900 bolts are respectively

$P(\text{def}) = 0.1$ mean =

$P(\text{Not def}) = 0.9$ S.D =

$n = 900$

$\text{Mean} = E[X] = np$

$= 900 \times 0.1 = 90$

$S.D = \sqrt{npq}$

$= \sqrt{900 \times 0.1 \times 0.9}$

$= \sqrt{81}$

$= 9$

$S.D = \sqrt{np(1-p)}$

A ✓ 90 and 9

B 9 and 90

C 81 and 9

D 9 and 81



THANK - YOU